

DEFI VAULT INFRASTRUCTURE

BotVault

The Premier ERC-4626 Native Liquidity Vault for the BOT Network

● Live on BOT Testnet (Chain ID 968)

OpenZeppelin 5.x • Inflation Protected (Offset 6) • Viem Powered

The Problem in Native Asset DeFi

Emerging Layer-1 ecosystems like BOT face severe friction and security vectors when utilizing native assets in modern decentralized finance.



UX Friction & Approvals

Users must wrap BOT to WBOT, approve allowances, deposit, and unwrap. Each extra transaction adds gas fees, latency, and drop-off rates.



Inflation & Donation Attacks

Empty or low-liquidity ERC-4626 vaults without decimal offsets can have share prices manipulated by donation attacks, stealing victim deposits.



Composability Deficit

Without a standardized ERC-4626 yield/share token, lending protocols, DEXs, and structured yield aggregators cannot easily integrate native BOT.

The Solution: BotVault

A battle-tested ERC-4626 vault with direct native BOT handling, institutional security, and automated inflation protection.



1-Click Native BOT Flow

Call `depositNative()` directly with native BOT to receive vBOT shares, and `redeemNative()` to exit to BOT. No approval transactions needed.



Permanent Inflation Shield

Overriding `_decimalsOffset() = 6` creates virtual shares and assets that make donation manipulation attacks mathematically unprofitable.



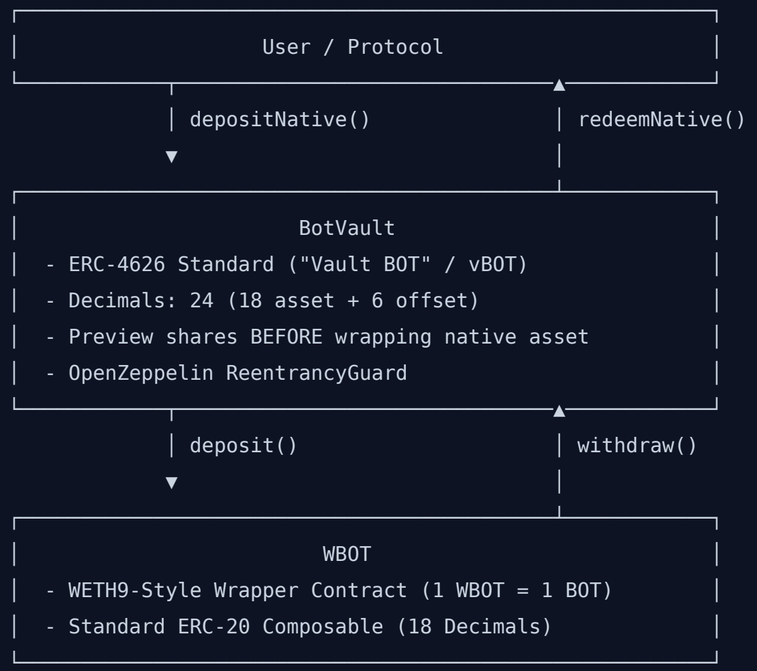
Checks-Effects-Interactions

Full OpenZeppelin `ReentrancyGuard` integration across all entry points. Safe unwrapping and native transfers prevent reentrancy exploits.

Key Result: Users get the seamless experience of native BOT deposits while protocols receive fully compliant, composable ERC-4626 vBOT shares.

Dual-Contract Clean Architecture

Modular separation between standard WETH9 asset wrapping and the ERC-4626 tokenized vault engine.



Precise Share Valuation

`shares = assets * (totalSupply + 10^6) / (totalAssets + 1)`
Round-trip deposit & redemption preserves 100% of user assets without precision loss.

Foundry Invariant Verification

Fuzzed across **128,000 calls** in Foundry: `totalAssets` strictly equals WBOT balance, and no actor can ever extract more than their fair share.

Live On-Chain Deployment & Traction

Deployed, verified, and battle-tested with multi-wallet production traffic on the BOT Testnet.

968	24	100%	128k
BOT Testnet Chain ID	Live User Test Transactions	Foundry Test Pass Rate	Invariant Fuzz Runs Passed

Verified Contracts on BohrScan

Verified Source Code

WBOT Address: 0x3dE55Ad41C1e13523d5A6Cc29a4794883D4aeF94 • scan.bohr.life/address/0x3dE...F94

BotVault Address: 0xb884156921A5508f9094a96C0848fad0Bd6c9Ff5 • scan.bohr.life/address/0xb88...Ff5

Modern, Minimalist Web Dashboard

Built with Vite, React 18, TypeScript, and viem. Zero bloated UI libraries for sub-second page loads.

Real-Time Stats Panel

Live RPC polling for Vault TVL, share price (BOT per vBOT), user's vBOT balance, and actual BOT net asset value.

1-Click Network Switch

Automatically detects wallet chain ID and provides a seamless "Switch to BOT Testnet" / "Add Network" prompt.

Transaction Tracking

Real-time status banners with animated feedback and direct clickable links to transaction hashes on BohrScan explorer.

Production Quality: Typesafe viem client, comprehensive error handling, responsive dark theme, and automated stats refresh after every transaction.

Market Opportunity & Strategic Roadmap

Positioned as the universal liquidity standard for the expanding BOHR blockchain ecosystem.

PHASE 1 (DELIVERED)

Foundational Layer

- OpenZeppelin 5.x contracts
- 128k invariant fuzz suite
- Testnet deployment & BohrScan verification
- Vite + React + viem frontend

PHASE 2 (Q1 2027)

Mainnet & Audits

- Third-party security audit certification
- BOT Mainnet contract deployment
- DEX liquidity pools (vBOT / BOT pairs)
- Money market collateral listing

PHASE 3 (Q2 2027)

Yield & Expansion

- Validator staking delegation strategy
- Auto-compounding yield mechanisms
- Institutional SDK & developer API
- Cross-chain vault bridges

GET STARTED TODAY

Unlocking Native BOT Liquidity

BotVault provides the battle-tested, trustless, and inflation-protected foundation for decentralized finance on BOT.

RPC: https://rpc.bohr.life

Explorer: https://scan.bohr.life

Chain ID: 968